

Question ID 95ca2683

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 95ca2683

In triangle ABC , the measure of angle A is 54° , the measure of angle B is 90° , and the measure of angle C is $\left(\frac{k}{2}\right)^\circ$. What is the value of k ?

- A. 36
- B. 45
- C. 72
- D. 108

ID: 95ca2683 Answer

Correct Answer:

C

Rationale

Choice C is correct. The sum of the interior angles of a triangle is 180° . It's given that the interior angles of triangle ABC are 54° , 90° , and $\left(\frac{k}{2}\right)^\circ$. It follows that $54 + 90 + \frac{k}{2} = 180$, or $144 + \frac{k}{2} = 180$. Subtracting 144 from each side of this equation yields $\frac{k}{2} = 36$. Multiplying each side of this equation by 2 yields $k = 72$. Therefore, the value of k is 72 .

Choice A is incorrect. This is the value of $\frac{k}{2}$, not k .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Medium

Question ID 4ff7b652

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 4ff7b652

Right triangles LMN and PQR are similar, where L and M correspond to P and Q , respectively. Angle M has a measure of 53° . What is the measure of angle Q ?

- A. 37°
- B. 53°
- C. 127°
- D. 143°

ID: 4ff7b652 Answer

Correct Answer:
B

Rationale

Choice B is correct. It's given that triangle LMN is similar to triangle PQR . Corresponding angles of similar triangles are congruent. Since angle M and angle Q correspond to each other, they must be congruent. Therefore, if the measure of angle M is 53° , then the measure of angle Q is also 53° .

Choice A is incorrect and may result from concluding that angle M and angle Q are complementary rather than congruent.

Choice C is incorrect and may result from concluding that angle M and angle Q are supplementary rather than congruent.

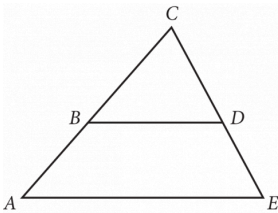
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID 6dd463ca

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 6dd463ca



Note: Figure not drawn to scale.

In the figure above, segments AE and BD are parallel. If angle BDC measures 58° and angle ACE measures 62° , what is the measure of angle CAE ?

- A. 58°
- B. 60°
- C. 62°
- D. 120°

ID: 6dd463ca Answer

Correct Answer:
B

Rationale

Choice B is correct. It’s given that angle ACE measures 62° . Since segments AE and BD are parallel, angles BDC and CEA are congruent. Therefore, angle CEA measures 58° . The sum of the measures of angles ACE , CEA , and CAE is 180° since the sum of the interior angles of triangle ACE is equal to 180° . Let the measure of angle CAE be x° . Therefore, $62 + 58 + x = 180$, which simplifies to $x = 60$. Thus, the measure of angle CAE is 60° .

Choice A is incorrect. This is the measure of angle AEC , not that of angle CAE . Choice C is incorrect. This is the measure of angle ACE , not that of CAE . Choice D is incorrect. This is the sum of the measures of angles ACE and CEA .

Question Difficulty:
Medium

Question ID 7a8ad237

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 7a8ad237

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 69° . What is the measure, in degrees, of angle F ?

ID: 7a8ad237 Answer

Correct Answer:
21

Rationale

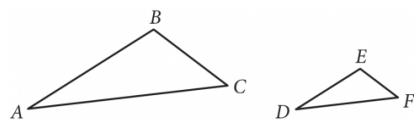
The correct answer is **21**. It's given that triangles ABC and DEF are congruent with angle A corresponding to angle D . Corresponding angles of congruent triangles are congruent and, therefore, have equal measure. It's given that the measure of angle A is 69° . It follows that the measure of angle D is also 69° . It's given that angle E is a right angle. Therefore, the measure of angle E is 90° . Let x represent the measure, in degrees, of angle F . Since the measures of the angles in a triangle sum to 180° , it follows that $69 + 90 + x = 180$, or $159 + x = 180$. Subtracting **159** from both sides of this equation yields $x = \mathbf{21}$. Therefore, the measure, in degrees, of angle F is **21**.

Question Difficulty:
Medium

Question ID 1c3d613c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 1c3d613c



Note: Figures not drawn to scale.

Triangle ABC and triangle DEF are shown. The relationship between the side lengths of the two triangles is such that $\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF} = 3$. If the measure of angle BAC is 20° , what is the measure, in degrees, of angle EDF ? (Disregard the degree symbol when gridding your answer.)

ID: 1c3d613c Answer

Rationale

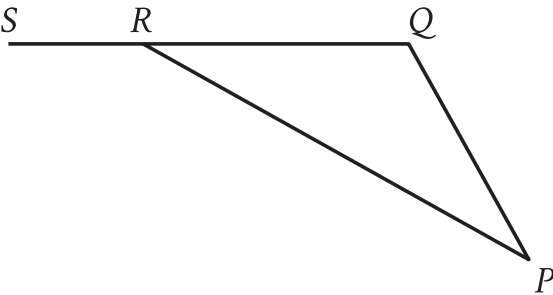
The correct answer is 20. By the equality given, the three pairs of corresponding sides of the two triangles are in the same proportion. By the side-side-side (SSS) similarity theorem, triangle ABC is similar to triangle DEF . In similar triangles, the measures of corresponding angles are congruent. Since angle BAC corresponds to angle EDF , these two angles are congruent and their measures are equal. It's given that the measure of angle BAC is 20° , so the measure of angle EDF is also 20° .

Question Difficulty:
Medium

Question ID 014edcb7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 014edcb7



Note: Figure not drawn to scale.

In triangle PQR , $\overline{QR}$ is extended to point S . The measure of $\angle PQR$ is 132° , and the measure of $\angle PRS$ is 163° . What is the measure of $\angle QPR$?

- A. 48°
- B. 31°
- C. 24°
- D. 17°

ID: 014edcb7 Answer

Correct Answer:

B

Rationale

Choice B is correct. In the figure shown, since $\overline{QS}$ is a line segment, the sum of the measures of $\angle PRS$ and $\angle PRQ$ is 180° . It's given that the measure of $\angle PRS$ is 163° . Thus, the measure of $\angle PRQ$ is $(180 - 163)^\circ$, or 17° . The sum of the measures of the interior angles of a triangle is 180° . It's given that the measure of $\angle PQR$ is 132° . Therefore, the measure of $\angle QPR$ is $(180 - 17 - 132)^\circ$, or 31° .

Choice A is incorrect. This is the measure of the supplement of $\angle PQR$, not the measure of $\angle QPR$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the measure of $\angle PRQ$, not the measure of $\angle QPR$.

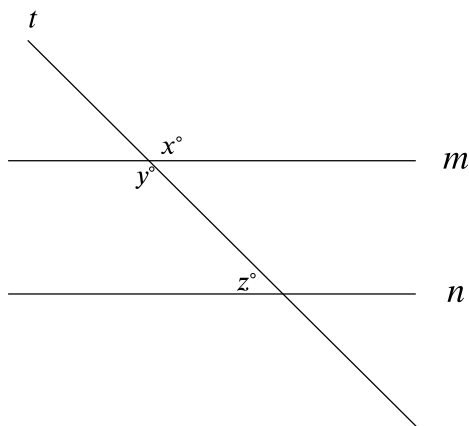
Question Difficulty:

Medium

Question ID 2adbf1b1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 2adbf1b1



Note: Figure not drawn to scale.

In the figure, lines m and n are parallel. If $x = 6k + 13$ and $y = 8k - 29$, what is the value of z ?

- A. 3
- B. 21
- C. 41
- D. 139

ID: 2adbf1b1 Answer

Correct Answer:
C

Rationale

Choice C is correct. Vertical angles, which are angles that are opposite each other when two lines intersect, are congruent. The figure shows that lines t and m intersect. It follows that the angle with measure x° and the angle with measure y° are vertical angles, so $x = y$. It's given that $x = 6k + 13$ and $y = 8k - 29$. Substituting $6k + 13$ for x and $8k - 29$ for y in the equation $x = y$ yields $6k + 13 = 8k - 29$. Subtracting $6k$ from both sides of this equation yields $13 = 2k - 29$. Adding 29 to both sides of this equation yields $42 = 2k$, or $2k = 42$. Dividing both sides of this equation by 2 yields $k = 21$. It's given that lines m and n are parallel, and the figure shows that lines m and n are intersected by a transversal, line t . If two parallel lines are intersected by a transversal, then the same-side interior angles are supplementary. It follows that the same-side interior angles with measures y° and z° are supplementary, so $y + z = 180$. Substituting $8k - 29$ for y in this equation yields $8k - 29 + z = 180$. Substituting 21 for k in this equation yields $8(21) - 29 + z = 180$, or $139 + z = 180$. Subtracting 139 from both sides of this equation yields $z = 41$. Therefore, the value of z is 41 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of k , not z .

Choice D is incorrect. This is the value of x or y , not z .

Question Difficulty:

Medium

Question ID 901e3285

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 901e3285

In triangle ABC , the measure of angle A is 50° . If triangle ABC is isosceles, which of the following is NOT a possible measure of angle B ?

- A. 50°
- B. 65°
- C. 80°
- D. 100°

ID: 901e3285 Answer

Correct Answer:
D

Rationale

Choice D is correct. The sum of the three interior angles in a triangle is 180° . It's given that angle A measures 50° . If angle B measured 100° , the measure of angle C would be $180^\circ - (50^\circ + 100^\circ) = 30^\circ$. Thus, the measures of the angles in the triangle would be 50° , 100° , and 30° . However, an isosceles triangle has two angles of equal measure. Therefore, angle B can't measure 100° .

Choice A is incorrect. If angle B has measure 50° , then angle C would measure $180^\circ - (50^\circ + 50^\circ) = 80^\circ$, and 50° , 50° , and 80° could be the angle measures of an isosceles triangle. Choice B is incorrect. If angle B has measure 65° , then angle C would measure $180^\circ - (65^\circ + 50^\circ) = 65^\circ$, and 50° , 65° , and 65° could be the angle measures of an isosceles triangle. Choice C is incorrect. If angle B has measure 80° , then angle C would measure $180^\circ - (80^\circ + 50^\circ) = 50^\circ$, and 50° , 80° , and 50° could be the angle measures of an isosceles triangle.

Question Difficulty:
Medium

Question ID c7bed21d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: c7bed21d

Quadrilateral $P'Q'R'S'$ is similar to quadrilateral $PQRS$, where P, Q, R , and S correspond to P', Q', R' , and S' , respectively. The measure of angle P is 30° , the measure of angle Q is 50° , and the measure of angle R is 70° . The length of each side of $P'Q'R'S'$ is 3 times the length of each corresponding side of $PQRS$. What is the measure of angle P' ?

- A. 10°
- B. 30°
- C. 40°
- D. 90°

ID: c7bed21d Answer

Correct Answer:

B

Rationale

Choice B is correct. It's given that quadrilateral $P'Q'R'S'$ is similar to quadrilateral $PQRS$, where P, Q, R , and S correspond to P', Q', R' , and S' , respectively. Since corresponding angles of similar quadrilaterals are congruent, it follows that the measure of angle P is equal to the measure of angle P' . It's given that the measure of angle P is 30° . Therefore, the measure of angle P' is 30° .

Choice A is incorrect. This is $\frac{1}{3}$ the measure of angle P' .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is 3 times the measure of angle P' .

Question Difficulty:

Medium

Question ID 2085e10e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 2085e10e

In triangle DEF , the measure of angle D is 47° and the measure of angle E is 97° . In triangle RST , the measure of angle R is 47° and the measure of angle S is 97° . Which of the following additional pieces of information is needed to determine whether triangle DEF is similar to triangle RST ?

- A. The measure of angle F
- B. The measure of angle T
- C. The measure of angle F and the measure of angle T
- D. No additional information is needed.

ID: 2085e10e Answer

Correct Answer:
D

Rationale

Choice D is correct. When two angles of one triangle are congruent to two angles of another triangle, the triangles are similar. It's given that in triangle DEF , the measure of angle D is 47° and the measure of angle E is 97° . It's also given that in triangle RST , the measure of angle R is 47° and the measure of angle S is 97° . It follows that angle D is congruent to angle R and that angle E is congruent to angle S . Therefore, triangle DEF is similar to triangle RST and no additional information is needed.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty:
Medium

Question ID fd8745fc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: fd8745fc

In triangle JKL , the measures of $\angle K$ and $\angle L$ are each 48° . What is the measure of $\angle J$, in degrees? (Disregard the degree symbol when entering your answer.)

ID: fd8745fc Answer

Correct Answer:

84

Rationale

The correct answer is **84**. The sum of the measures of the interior angles of a triangle is 180° . It's given that in triangle JKL , the measures of $\angle K$ and $\angle L$ are each 48° . Adding the measures, in degrees, of $\angle K$ and $\angle L$ gives $48 + 48$, or **96**. Therefore, the measure of $\angle J$, in degrees, is $180 - 96$, or **84**.

Question Difficulty:

Medium

Question ID d3fe472f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: d3fe472f

Triangle ABC is similar to triangle XYZ , such that A , B , and C correspond to X , Y , and Z respectively. The length of each side of triangle XYZ is 2 times the length of its corresponding side in triangle ABC . The measure of side AB is 16 . What is the measure of side XY ?

- A. 14
- B. 16
- C. 18
- D. 32

ID: d3fe472f Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that triangle ABC is similar to triangle XYZ , such that A , B , and C correspond to X , Y , and Z , respectively. Therefore, side AB corresponds to side XY . Since the length of each side of triangle XYZ is 2 times the length of its corresponding side in triangle ABC , it follows that the measure of side XY is 2 times the measure of side AB . Thus, since the measure of side AB is 16 , then the measure of side XY is $2(16)$, or 32 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the measure of side AB , not side XY .

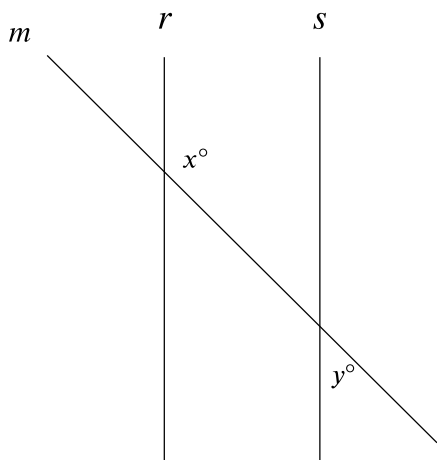
Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID a4c05a1b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: a4c05a1b



Note: Figure not drawn to scale.

In the figure shown, lines r and s are parallel, and line m intersects both lines. If $y < 65$, which of the following must be true?

- A. $x < 115$
- B. $x > 115$
- C. $x + y < 180$
- D. $x + y > 180$

ID: a4c05a1b Answer

Correct Answer:

B

Rationale

Choice B is correct. In the figure shown, the angle measuring y° is congruent to its vertical angle formed by lines s and m , so the measure of the vertical angle is also y° . The vertical angle forms a same-side interior angle pair with the angle measuring x° . It's given that lines r and s are parallel. Therefore, same-side interior angles in the figure are supplementary, which means the sum of the measure of the vertical angle and the measure of the angle measuring x° is 180° , or $x + y = 180$. Subtracting x from both sides of this equation yields $y = 180 - x$. Substituting $180 - x$ for y in the inequality $y < 65$ yields $180 - x < 65$. Adding x to both sides of this inequality yields $180 < 65 + x$. Subtracting 65 from both sides of this inequality yields $115 < x$, or $x > 115$. Thus, if $y < 65$, it must be true that $x > 115$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. $x + y$ must be equal to, not less than, 180 .

Choice D is incorrect. $x + y$ must be equal to, not greater than, 180 .

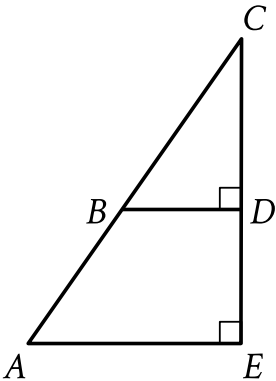
Question Difficulty:

Medium

Question ID 2f7c92ad

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 2f7c92ad



Note: Figure not drawn to scale.

In the figure shown, triangle CAE is similar to triangle CBD . The measure of angle CBD is 57° , and $AE = 26(BD)$. What is the measure of angle CAE ?

- A. $(26 \cdot 57)^\circ$
- B. $(26 + 57)^\circ$
- C. 57°
- D. 26°

ID: 2f7c92ad Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that triangle CAE is similar to triangle CBD . Corresponding angles in similar triangles have equal measure. Angle BCD and angle ACE represent the same angle. It follows that angle BCD and angle ACE have equal measure and are corresponding angles. It's given in the figure that angle BDC and angle AEC are right angles and therefore have equal measure. It follows that angle BDC and angle AEC are corresponding angles. Therefore, angle CBD and angle CAE are corresponding angles and have equal measure. It's given that the measure of angle CBD is 57° , so the measure of angle CAE is 57° .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

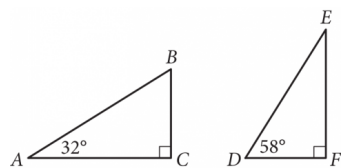
Question Difficulty:

Medium

Question ID 933fee1a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 933fee1a



Triangles ABC and DEF are shown above. Which of the

following is equal to the ratio $\frac{BC}{AB}$?

- A. $\frac{DE}{DF}$
- B. $\frac{DF}{DE}$
- C. $\frac{DF}{EF}$
- D. $\frac{EF}{DE}$

ID: 933fee1a Answer

Correct Answer:
B

Rationale

Choice B is correct. In right triangle ABC , the measure of angle B must be 58° because the sum of the measure of angle A , which is 32° , and the measure of angle B is 90° . Angle D in the right triangle DEF has measure 58° . Hence, triangles ABC and DEF are similar (by angle-angle similarity). Since $\overline{BC}$ is the side opposite to the angle with measure 32° and AB is the hypotenuse in right triangle ABC , the ratio $\frac{BC}{AB}$ is equal to $\frac{DF}{DE}$.

Alternate approach: The trigonometric ratios can be used to answer this question. In right triangle ABC , the ratio $\frac{BC}{AB} = \sin(32^\circ)$. The angle E in triangle DEF has measure 32° because $m(\angle D) + m(\angle E) = 90^\circ$. In triangle DEF , the ratio $\frac{DF}{DE} = \sin(32^\circ)$. Therefore, $\frac{DF}{DE} = \frac{BC}{AB}$.

Choice A is incorrect because $\frac{DE}{DF}$ is the reciprocal of the ratio $\frac{BC}{AB}$. Choice C is incorrect because $\frac{DF}{EF} = \frac{BC}{AC}$, not $\frac{BC}{AB}$.

Choice D is incorrect because $\frac{EF}{DE} = \frac{AC}{AB}$, not $\frac{BC}{AB}$.

Question Difficulty:

Medium

Question ID d7a8aa9c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: d7a8aa9c

Each side of equilateral triangle S is multiplied by a scale factor of k to create equilateral triangle T. The length of each side of triangle T is greater than the length of each side of triangle S. Which of the following could be the value of k ?

- A. $\frac{29}{28}$
- B. 1
- C. $\frac{28}{29}$
- D. 0

ID: d7a8aa9c Answer

Correct Answer:
A

Rationale

Choice A is correct. It's given that each side of equilateral triangle S is multiplied by a scale factor of k to create equilateral triangle T. Since the length of each side of triangle T is greater than the length of each side of triangle S, the scale factor of k must be greater than 1. Of the given choices, only $\frac{29}{28}$ is greater than 1.

Choice B is incorrect. If each side of equilateral triangle S is multiplied by a scale factor of 1, the length of each side of triangle T would be equal to the length of each side of triangle S.

Choice C is incorrect. If each side of equilateral triangle S is multiplied by a scale factor of $\frac{28}{29}$, the length of each side of triangle T would be less than the length of each side of triangle S.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID 055aafe7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 055aafe7

Triangle ABC is similar to triangle XYZ , where A , B , and C correspond to X , Y , and Z , respectively. In triangle ABC , the length of $\overline{AB}$ is 170 and the length of $\overline{BC}$ is 850. In triangle XYZ , the length of $\overline{YZ}$ is 60. What is the length of $\overline{XY}$?

- A. 204
- B. 182
- C. 60
- D. 12

ID: 055aafe7 Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that triangle ABC is similar to triangle XYZ , where A , B , and C correspond to X , Y , and Z , respectively. It follows that side AB corresponds to side XY and side BC corresponds to side YZ . Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{XY}{AB} = \frac{YZ}{BC}$. Substituting 170 for AB , 60 for YZ , and 850 for BC in this equation yields $\frac{XY}{170} = \frac{60}{850}$. Multiplying each side of this equation by 170 yields $XY = 12$. Therefore, the length of $\overline{XY}$ is 12.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

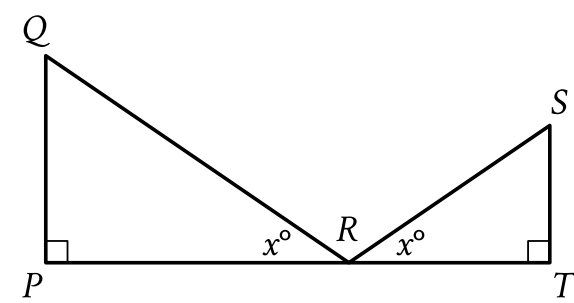
Choice C is incorrect. This is the length of $\overline{YZ}$, not $\overline{XY}$.

Question Difficulty:
Medium

Question ID 51f26ce8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 51f26ce8



Note: Figure not drawn to scale.

$\triangle QPR$ is similar to $\triangle STR$. The lengths represented by $\overline{ST}$, $\overline{QP}$, $\overline{PR}$, and $\overline{QR}$ in the figure are **14**, **15**, **20**, and **25**, respectively. What is the length of $\overline{SR}$?

- A. $\frac{350}{15}$
- B. $\frac{350}{20}$
- C. $\frac{210}{20}$
- D. $\frac{210}{25}$

ID: 51f26ce8 Answer

Correct Answer:

A

Rationale

Choice A is correct. The figure shows that angle P in $\triangle QPR$ and angle T in $\triangle STR$ are right angles. It follows that angle P is congruent to angle T . The figure also shows that the measures of angle QRP and angle SRT are both x° . Therefore, angle QRP is congruent to angle SRT . It's given that $\triangle QPR$ is similar to $\triangle STR$. Since angle P is congruent to angle T , and angle QRP is congruent to angle SRT , it follows that $\overline{QR}$ corresponds to $\overline{SR}$, and $\overline{QP}$ corresponds to $\overline{ST}$. Since corresponding sides of similar triangles are proportional, it follows that $\frac{SR}{QR} = \frac{ST}{QP}$. It's also given that the lengths of $\overline{ST}$, $\overline{QP}$, and $\overline{QR}$ are **14**, **15**, and **25**, respectively. Substituting **14** for $\overline{ST}$, **15** for $\overline{QP}$, and **25** for $\overline{QR}$ in the equation $\frac{SR}{QR} = \frac{ST}{QP}$ yields $\frac{SR}{25} = \frac{14}{15}$. Multiplying each side of this equation by **25** yields $SR = (\frac{14}{15})(25)$, or $SR = \frac{350}{15}$. Thus, the length of $\overline{SR}$ is $\frac{350}{15}$.

Choice B is incorrect. This is the result of solving the equation $\frac{SR}{25} = \frac{14}{20}$, not $\frac{SR}{25} = \frac{14}{15}$.

Choice C is incorrect. This is the result of solving the equation $\frac{SR}{14} = \frac{15}{20}$, not $\frac{SR}{25} = \frac{14}{15}$.

Choice D is incorrect. This is the result of solving the equation $\frac{SR}{14} = \frac{15}{25}$, not $\frac{SR}{25} = \frac{14}{15}$.

Question Difficulty:

Medium

Question ID 94364a79

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 94364a79

Two nearby trees are perpendicular to the ground, which is flat. One of these trees is **10** feet tall and has a shadow that is **5** feet long. At the same time, the shadow of the other tree is **2** feet long. How tall, in feet, is the other tree?

- A. **3**
- B. **4**
- C. **8**
- D. **27**

ID: 94364a79 Answer

Correct Answer:
B

Rationale

Choice B is correct. Each tree and its shadow can be modeled using a right triangle, where the height of the tree and the length of its shadow are the legs of the triangle. At a given point in time, the right triangles formed by two nearby trees and their respective shadows will be similar. Therefore, if the height of the other tree is x , in feet, the value of x can be calculated by solving the proportional relationship $\frac{10 \text{ feet tall}}{5 \text{ feet long}} = \frac{x \text{ feet tall}}{2 \text{ feet long}}$. This equation is equivalent to $\frac{10}{5} = \frac{x}{2}$, or $2 = \frac{x}{2}$. Multiplying each side of the equation $2 = \frac{x}{2}$ by 2 yields $4 = x$. Therefore, the other tree is **4 feet** tall.

Choice A is incorrect and may result from calculating the difference between the lengths of the shadows, rather than the height of the other tree.

Choice C is incorrect and may result from calculating the difference between the height of the **10**-foot-tall tree and the length of the shadow of the other tree, rather than calculating the height of the other tree.

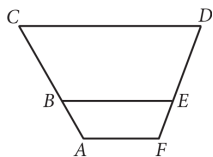
Choice D is incorrect and may result from a conceptual or calculation error.

Question Difficulty:
Medium

Question ID 81b664bc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 81b664bc



In the figure above, $\overline{AF}$, $\overline{BE}$, and $\overline{CD}$ are parallel. Points B and E lie on $\overline{AC}$ and $\overline{FD}$, respectively. If $AB = 9$, $BC = 18.5$, and $FE = 8.5$, what is the length of $\overline{ED}$, to the nearest tenth?

- A. 16.8
- B. 17.5
- C. 18.4
- D. 19.6

ID: 81b664bc Answer

Correct Answer:
B

Rationale

Choice B is correct. Since $\overline{AF}$, $\overline{BE}$, and $\overline{CD}$ are parallel, quadrilaterals $AFEB$ and $BEDC$ are similar. Let x represent the length of $\overline{ED}$. With similar figures, the ratios of the lengths of corresponding sides are equal. It follows that $\frac{9}{18.5} = \frac{8.5}{x}$. Multiplying both sides of this equation by 18.5 and by x yields $9x = (18.5)(8.5)$, or $9x = 157.25$. Dividing both sides of this equation by 9 yields $x = 17.47$, which to the nearest tenth is 17.5.

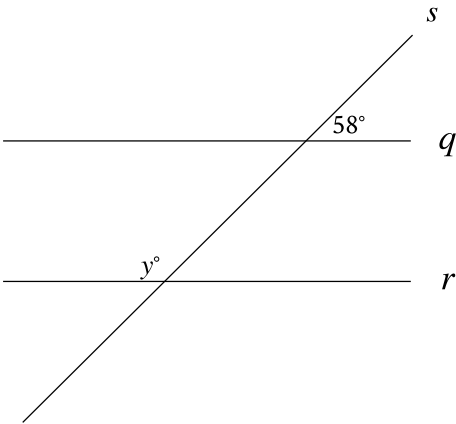
Choices A, C, and D are incorrect and may result from errors made when setting up the proportion.

Question Difficulty:
Medium

Question ID 686b5212

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 686b5212



Note: Figure not drawn to scale.

In the figure, line q is parallel to line r , and both lines are intersected by line s . If $y = 2x + 8$, what is the value of x ?

ID: 686b5212 Answer

Correct Answer:
57

Rationale

The correct answer is **57**. Based on the figure, the angle with measure y° and the angle vertical to the angle with measure 58° are same side interior angles. Since vertical angles are congruent, the angle vertical to the angle with measure 58° also has measure 58° . It's given that lines q and r are parallel. Therefore, same side interior angles between lines q and r are supplementary. It follows that $y + 58 = 180$. If $y = 2x + 8$, then the value of x can be found by substituting $2x + 8$ for y in the equation $y + 58 = 180$, which yields $(2x + 8) + 58 = 180$, or $2x + 66 = 180$. Subtracting **66** from both sides of this equation yields $2x = 114$. Dividing both sides of this equation by **2** yields $x = 57$. Thus, if $y = 2x + 8$, the value of x is **57**.

Question Difficulty:
Medium